

U.G.C. NET Exam

COMPUTER

(Solved with Explanation)

1. **Eco system is a Frame work for**
 (a) Building a Computer System
 (b) Building Internet Market
 (c) Building Offline Market
 (d) Building Market

Ans. (b) : Eco system is a Frame work for building internet market means once you built content, it needs to be shared across the web so it will done by building internet market.

2. **The efficiency (E) and speed up (sp) for Multiprocessor with p processors satisfies**
 (a) $E \leq p$ and $s_p \leq p$
 (b) $E \leq 1$ and $s_p \leq p$
 (c) $E \leq p$ and $s_p \leq 1$
 (d) $E \leq 1$ and $s_p \leq 1$

Ans. (b) : Efficiency (E) can be 0 to 1 or not more than 0 to 100%. even 1 is just theoretical speed up for multiprocessor. Speed of multiprocessor S_p can never exceed (p) number of processors.
 So answer is (b).

3. **Match the following :**

List-I

- A. Critical region
 B. Wait/signal
 C. Working set
 D. Dead lock

List-II

1. Hoares Monitor
 2. Mutual exclusion
 3. Principal of locality
 4. Circular wait

Codes :

	A	B	C	D
(a)	2	1	3	4
(b)	1	2	4	3
(c)	2	3	1	4
(d)	1	3	2	4

Ans. (a) :

(a) Critical region—Critical region is the region of values that corresponds to the rejection of the null hypothesis at some chosen probability level. So it satisfies, (ii) Mutual exclusion

(b) Wait/Signal holds Hoares Monitor

(c) Working Set—It directly address thrashing problem in which is the set of pages referenced by the process during the most recent w page reference is the principle of locality.

(d) Dead lock—It hold circular wait because circular wait is a condition of dead lock.

4. **The technique of temporarily delaying outgoing acknowledgements so that they can be hooked onto the next outgoing data frame is known as**

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- (a) Bit stuffing (b) Piggy backing
(c) Pipelining (d) Broadcasting

Ans. (b) : The technique of temporarily delaying outgoing acknowledgments, so that they can be hooked onto the next outgoing data frame is known as piggy backing.

5. _____ is process of extracting previously non known valid and actionable information from large data to make crucial business and strategic decisions.

- (a) Data Management (b) Data base
(c) Data Mining (d) Meta Data

Ans. (c) : Data mining is defined as extracting information from huge sets of data, we can say that mining is the procedure of mining knowledge of data.

6. The aspect ratio of an image is defined as

- (a) The ratio of width to its height measured in unit length.
(b) The ratio of height to width measured in number of pixels.
(c) The ratio of depth to width measured in unit length
(d) The ratio of width to depth measured in number of pixels.

Ans. (a) : Because the aspect ratio of an image describes the proportional relationship between its width and height. It is commonly expressed as two number separated by a color, as in 16:9.

7. Which of the following features will characterize an OS as multi-programmed OS?

- (A) More than one program may be loaded into main memory at the same time.
(B) If a program waits for certain event another program is immediately scheduled.
(C) If the execution of a program terminates, another program is immediately scheduled.
(a) (A) only
(b) (A) and (B) only
(c) (A) and (C) only
(d) (A), (B) and (C) only

Ans. (d) : A Multiprogramming system permits multiple programs to be loaded into memory and execute the programs concurrently. Concurrent execution of programs results into improved system throughput and resource, utilization. This potential is realized by a class of operating system among a multiple of active programs. Such operating systems usually have the prefix multi in their names, such as multitasking or multiprogramming.

8. Using RSA algorithm, what is the value of cipher text C, if the plain text M = 5 and p = 3, q = 11 & d = 7 ?

- (a) 33 (b) 5
(c) 25 (d) 26

Ans. (d) :

Given P = 3, Q = 11, n = (p × q) = 3 × 11 = 33

m = (P-1) × (Q-1) = (2 × 10) = 20

find a small odd integer, that is relatively prime to M.

If e = 3, then GCD (3, 20) = 1. e should be small & prime & So we let e = 3, d is given, d = 7

public key = (e, n). (values of e and n are known)

To encrypt a message we apply the public key to the function $E(s) = S^e \pmod n$

Where S is the given message and e and n represents the public key integer pair. In the above question, the plain text M = 5 plain text needs to be encrypted using above formula

$$= 5^3 \pmod{33}$$

$$= 125 \pmod{33}$$

$$= 26$$

9. You are given an OR problem and a XOR problem to solve. Then, which one of the following statements is true?

- (a) Both OR and XOR problems can be solved using single layer perception.
(b) OR problem can be solved using single layer perception and XOR problem can be solved using self organizing maps.
(c) OR problem can be solved using radial basis function and XOR problem can be solved using single layer perception.
(d) OR problem can be solved using single layer perception and XOR problem can be solved using radial basis function.

Ans. (d) : OR problem can be solved using single layer perception and XOR problem can be solved using radial basis function.

10. Match the following :

	List-I	List-II
a.	Application layer	1. TCP
b.	Transport layer	2. HDLC
c.	Network layer	3. HTTP
d.	Data layer	4. BGP

Codes :

	a	b	c	d
(a)	2	1	4	3
(b)	3	4	1	2
(c)	3	1	4	2
(d)	2	4	1	3

Ans. (c) :

(a) **Application layer**—It is the top most layer in OSI & TCP/IP layered model & when we use a web browser, which is actually using Hyper Text Transfer

Protocol (HTTP) to interact with the Network. HTTP is an application layer protocol.

(b) Transport layer—It offers peer to peer & end to end connection between two processes on remote hosts. It have TCP (Transmission Control Protocol) & UDP (User Datagram Protocol).

(c) Network layer—It takes the responsibility for routing packets from source to destination within or outside a subnet. So it refers BGP (Border Gateway Protocol)

(d) Data link layer—It hides the details of underlying H/W and represents itself to upper layer as the medium to communicate. So it refers HDLC (High Level Data Link Control).

11. The time complexities of some standard graph algorithms are given. Match each algorithm with its time complexity? (n and m are no. of nodes and edges respectively)

- | | |
|-----------------------------|------------------|
| a. Bellman Ford algorithm | 1. $O(m \log n)$ |
| b. Kruskals algorithm | 2. $O(n^3)$ |
| c. Floyd Wrashall algorithm | 3. $O(mn)$ |
| d. Topological sorting | 4. $O(n + m)$ |

Codes :

	a	b	c	d
(a)	3	1	2	4
(b)	2	4	3	1
(c)	3	4	1	2
(d)	2	1	3	4

Ans. (a) :

(a) Bellman Ford algorithm time complexity is $O(mn)$

(b) Kruskals algorithm time complexity is $O(m \log n)$

(c) Floyd Wrashall algorithm time complexity is $O(n^3)$

(d) Topological sorting time complexity is $O(n + m)$

Where n & m are number of nodes and edges respectively.

12. Let $V_1 = 2I - J + K$ and $V_2 = I + J - K$, then the angle between V_1 & V_2 and a vector perpendicular to both V_1 & V_2 shall be :

- (a) 90° and $(-2I + J - 3K)$
 (b) 60° and $(2I + J + 3K)$
 (c) 90° and $(2I + J - 3K)$
 (d) 90° and $(-2I - J + 3K)$

Ans. (d) : $V_1 \cdot V_2 = (2I - J + K) \cdot (I + J - K)$
 $= 2 - 1 - 1 = 0$

Hence angle between V_1 and V_2 is 90°

$$V_1 \times V_2 = \begin{vmatrix} I & K & J \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= I(-1-1) - J(-2-1) + K(2+1)$$

$$= -2I - J + 3K$$

So answer will be (b).

13. Consider a fuzzy set A defined on the interval $X = [0, 10]$ of integers by the membership Junction

$$\mu_A(x) = \frac{x}{x+2}$$

Then the α cut corresponding to $\alpha = 0.5$ will be

- (a) $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- (b) $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- (c) $\{2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- (d) $\{ \}$

Ans. (c) : Because, putting value of X from 0 to 10 in membership function $\mu_A(x) = x / x + 2$ we get
 $0/2, 1/3, 2/4, 3/5, 4/6, 5/7, 6/8, 7/9, 8/10, 9/11, 10/12$
means they are belonging with degree 0, 0.33, 0.5, 0.6
----- 0.83.

here α -cut = 0.5 so first two elements will not be included in the result as their degree of belongingness 0.5. So answer will be (c).

14. Let $T(n)$ be the function defined by $T(n) = 1$ and $T(n) = 2T(n/2) + \sqrt{n}$, which of the following is TRUE?

- (a) $T(n) = O(\sqrt{n})$
- (b) $T(n) = O(\log_2 n)$
- (c) $T(n) = O(n)$
- (d) $T(n) = O(n^2)$

Ans. (c): $T(n) = O(n^2)$

Because, when you apply master theorem on the function $T(n)$ then you get $T(n) = O(n)$ so answer will be (c).

15. In classful addressing, an IP address 123. 23. 156.4 belongs to class format.

- (a) A
- (b) B
- (c) C
- (d) D

Ans. (a) :

Because, $0 - 126 = \text{Class A}$

$128 - 191 = \text{Class B}$

$192 - 223 = \text{Class C}$

$224 - 239 = \text{Class D}$

$240 - 255 = \text{Class E}$

There, given address 123.23.256.4 belongs to class A

So answer will be (a).

16. The Mandelbrot set used for the construction of beautiful images is based on the following transformation:

$$x_{n+1} = x_n^2 + z$$

Here,

- (a) Both x & z are real numbers.
- (b) Both x & z are complex numbers.
- (c) x is real & z is complex.
- (d) x is complex & z is real.

Ans. (b) : Both x & z are complex numbers

The Mandelbrot set is the set of values of c in the complex plane for which the orbit of 0 under iteration

of the complex quadratic polynomial $z_{n+1} = z_n^2 + c$ remains bounded.

That is, a complex number c is part of the Mandelbrot set if, when starting with $z_0 = 0$ and applying the iteration repeatedly, the absolute value of z_n remains bounded however large n gets. So answer will be (b)

- 17. Which of the following permutations can be obtained in the output using a stack of size 3 elements assuming that input, sequence is 1, 2, 3, 4, 5 ?**

- (a) 3, 2, 1, 5, 4 (b) 5, 4, 3, 2, 1
(c) 3, 4, 5, 2, 1 (d) 3, 4, 5, 1, 2

Ans. (c)

Operation	P u s h (1)	P u s h (2)	P u s h (3)	P o p	P u s h (4)	P o p	P u s h (5)	P o p	P o p	P o p
		2	2	2	4	2	5	2		
Output	1	1	1	1 3	1 3	1 34	1 34	1 345	1 352	
									34521	

So option (c) is correct.

- 18. In a Linear Programming Problem, suppose there are 3 basic variables and 2 non-basic variables, then the possible number of basic solutions are**

- (a) 6 (b) 8
(c) 10 (d) 12

Ans. (c) : Because, Total number of basic solution are given by the equation

$$n! / m! * (n-m)!$$

where $m = 3$ no. of basic variables & $n = 3 + 2 = 5$ total no. of variable.

hence total solution = $5! / 3! 2! = 5 \times 4/2 = 10$

So answer is (c).

- 19. Identify the following activation function :**

$$\phi(V) = Z + \frac{1}{1 + \exp(-x * V + Y)},$$

Z, X, Y are parameters

- (a) Step function (b) Ramp function
(c) Sigmoid function (d) Gaussian function

Ans. (c) : A Sigmoid function is a mathematical function having an 'S' shape (Sigmoid-curve). A sigmoid function is a bounded differentiable real function that is defined for all real input values and has a positive everywhere.

- 20. The no. of ways to distribute n distinguishable objects into k distinguishable boxes, so that n_1 objects are placed into box i, $i = 1, 2, \dots, k$ equals which of the following?**

- (a) $\frac{n!}{n_1! + n_2! + \dots + n_k!}$
(b) $\frac{n_1! + n_2! + \dots + n_k!}{n_1! + n_2! n_3! + \dots + n_k!}$

$$(c) \frac{n!}{n_1! n_2! n_3! \dots n_k!}$$

$$(d) \frac{n_1! n_2! \dots + n_k!}{n_1! - n_2! - n_3! \dots - n_k!}$$

Ans. (c) : Because, the number of ways to distribute n distinguishable objects into K distinct boxes so that n_i objects are placed in box i , $i = 1, \dots, k$ & $n_1 + \dots + n_k = n$ is $\frac{n!}{n_1! n_2! n_3! \dots n_k!}$.

21. How many solutions do the following equations have?

$$x_1 + x_2 + x_3 = 11$$

$$\text{where } x_1 \geq 1, x_2 \geq 2, x_3 \geq 3$$

- (a) $C(7, 11)$ (b) $C(11, 3)$
(c) $C(14, 11)$ (d) $C(7, 5)$

Ans. (d) : Given, equation is: $x_1 + x_2 + x_3 = 11$

$$\text{where } x_1 \geq 1, x_2 \geq 2, x_3 \geq 3$$

when we satisfied given conditions, then

$$x_1 + x_2 + x_3 = 11 - (1 + 2 + 3) = 5$$

$$x_1 + x_2 + x_3 = 5$$

hence, number of possible solutions are:

$${}^{(n-1)+k}C_k = {}^{(3-1)+5}C_5 = {}^7C_5 = {}^7C_2$$

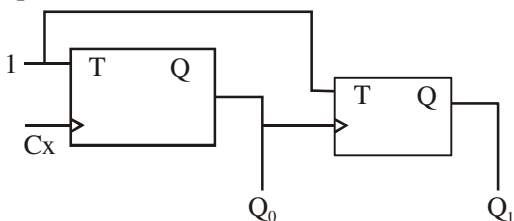
22. Which provides an interface to the TCP/IP suite protocols in Windows 95 and Windows NT?

- (a) FTP Active-X Control
(b) TCP/IP Active-X Control
(c) Calinsock Active-X Control
(d) HTML Active-X Control

Ans. (c) : Calinsock Active-X Control

Because, the window Sockets API (WSA), which was later shortened to winsock, is a technical specification that defines how windows network software should access network services. It defines a standard interface between a windows TCP/IP client application and the underlying TCP/IP protocol stack. The nomenclature is based on the Berkeley Sockets API model used in BSD for communication between programs.

23. What are the final values of Q_1 and Q_0 after 4 clock cycles, if initial values are 00 in the sequential circuit shown below :



- (a) 11 (b) 10
(c) 01 (d) 00

Ans. (d) :

T	Q_{n+1}
0	Q_n
1	Q_n

T_0	Q_0	T_1	Q_1
Initially	0		0
1	1	1	1
1	0	1	1
1	1	1	0
1	0	1	0

Therefore, $Q_1 Q_0$ sequence is:

$00_{\text{initial}} \rightarrow 11_{1\text{st}} \rightarrow 10_{2\text{nd}} \rightarrow 01_{3\text{rd}} \rightarrow 00_{4\text{th}}$

24. If dual has an unbounded solution, then its corresponding primal has

- (a) no feasible solution
- (b) unbounded solution
- (c) feasible solution
- (d) none of these

Ans. (a) : no feasible solution

Because, few property of primal & dual

- (1) The dual of dual linear programming problem is again the primal solution.
- (2) If either the primal or dual problem has unbounded solution the other problem has no feasible solution.
- (3) If either the primal has a finite optimal solution the other one also possesses the same & the optimal value of the objective function of 2 problems are same.

So answer is (a).

25. The number of distinct bracelets of five beads made up of red, blue, and green beads (two bracelets are indistinguishable if the rotation of one yield another) is,

- (a) 243
- (b) 81
- (c) 51
- (d) 47

Ans. (c) Verify that any necklace may be characterized via one of the patterns below.

Recall that there are $\frac{3!}{(3-n)!}$ ways to put 3 things into n ordered slots.

$$xxxxx : 3 = \frac{3!}{(3-1)!}$$

$$xxxxy : 6 = \frac{3!}{(3-2)!}$$

$$xxxyy : 6 = \frac{3!}{(3-2)!}$$

$$xxxyz : 6 = \frac{3!}{(3-3)!}$$

$$xxyyz : 6 = \frac{3!}{(3-3)!}$$

$$xxyxy : 6 = \frac{3!}{(3-2)!}$$

$$xxyxz : 6 = \frac{3!}{(3-3)!}$$

$$xxyzy : 6 = \frac{3!}{(3-3)!}$$

$$xyzyz : 6 = \frac{3!}{(3-2)!}$$

$$3+6+6+6+6+6+6+6+6 = 51$$

26. Which are the classifications of data used in Mobile Applications?

- (a) Private data, User data, Shared data.
- (b) Public data, User data, Virtual data.
- (c) Private data, Public data, Shared data.
- (d) Public data, Virtual data, User data.

Ans. (c) : private data are those data that are not made available to the general public such as password and accounts. **Public data** are those data that are for the public good. **Shared data** are those data that are a set amount of data for all smart phones user on a specific plan. Shared data plans include both text message & calling.

27. In an enhancement of a CPU design, the speed of a floating point unit has been increased by 20% and the speed of a fixed point unit has been increased by 10%. What is the overall speed achieved if the ratio of the number of floating point operations to the number of fixed point operations is 2 : 3 and the floating point operation used to take twice the time taken by the fixed point operation in original design?

- (a) 1.2
- (b) 1.55
- (c) 1.85
- (d) 1.285

Ans. (b) : Because, speed up = original time taken/new time taken

Let x be the time for a fixed point operation original time taken = $(3x + 2*2x) / 5 = 7x / 5$

New time taken = $((3x / 1.1) + 4x / 1.2) / 5 = 8x / 1.32 * 5$

So speed up = $7 * 1.32 / 8 = 1.155$

28. The initial basic feasible solution to the following transportation problem using Vogel's approximation method is

	D ₁	D ₂	D ₃	D ₄	Supply
S ₁	1	2	1	4	30
S ₂	3	3	2	1	50
S ₃	4	2	5	9	20
Demand	20	40	30	10	

- (a) $x_{11} = 20, x_{13} = 10, x_{21} = 20,$
 $x_{23} = 20, x_{24} = 10, x_{32} = 10,$
 Total cost = 180

- (b) $x_{11} = 20, x_{12} = 20, x_{13} = 10,$
 $x_{22} = 20, x_{23} = 20, x_{24} = 10,$
 Total cost = 180
- (c) $x_{11} = 20, x_{13} = 10, x_{22} = 20,$
 $x_{23} = 20, x_{24} = 10, x_{32} = 10,$
 Total cost = 180
- (d) None of the above

Ans. (c) :

	D₁	D₂	D₃	D₄	Supply
S₁	1(20)	2(10)	1	4	30
S₂	3	3(10)	2(30)	1(10)	50
S₃	4	2(20)	5	9	20
Demand	20	40	30	10	

The minimum total transportation cost is=
 $1 \times 20 + 2 \times 10 + 3 \times 10 + 2 \times 30 + 1 \times 10 + 2 \times 20 = 180$
 So answer is (c).

- 29. 58 lamps are to be connected to a single electric outlet by using an extension board each of which has four outlets the number of extension boards needed to connect all the light is**
- (a) 29 (b) 28
 (c) 20 (d) 19

Ans. (d) : Because, Connect first 4 lamps (L_1, L_2, L_3, L_4) to extension 1 & take power from extension 2. In extension 2 we can connect only 3 lamps because out of 4 one port is busy to supply power to extension 1. Similarly, for other extension we can connect only 3 lamps because one port is busy to supply power to other extension.

We can connect 4 lamps to first extension.
 number of extension required for remaining 54 lamps = 18 extension.

Total extension required to connect all 58 lamps = 18 + 1 = 19

- 30. Match the following with respect to the Mobile Computing Architecture.**

List-I

List-II

- | | |
|--|---------------------------------|
| a. Downlink control | 1. 100 Mbps |
| b. Radio communication data rate | 2. Residency latency (RL) |
| c. The average duration of user's stay in cell | 3. Sending data from a BS to MD |
| d. FDDI bandwidth | 4. 2-Mbps |

Codes :

- | | a | b | c | d |
|-----|----------|----------|----------|----------|
| (a) | 2 | 1 | 4 | 3 |
| (b) | 3 | 4 | 2 | 1 |
| (c) | 4 | 1 | 2 | 1 |
| (d) | 4 | 3 | 1 | 2 |

Ans. (b) : Because,

- (1) Downlink control is sending a data from BS to MD. Use of Downlink Control in Mobile computing Architecture: The base station subsystem (BSS) is the section of a traditional cellular telephones network which is responsible for handling traffic & signaling between a Mobile Device & the network switching subsystem.
- (b) Data Communication data rate is 2 Mbps
- (c) User are highly mobile and randomly enter and exit from cells. There is a parameter called Residence Latency (RL), which characterizes the average duration of a user's stay in the cell.
- (d) FDDI (Fiber Distributed Data Interfere) specifies 100 mbps token passing, dual-ring LAN using fiber-optic cable.

31. Which of the following flags are set when 'JMP' instruction is executed?

- (a) SF and CF
- (b) AF and CF
- (c) All flags
- (d) No flag is set

Ans. (d) : The JMP is instruction transfers extension to the address generated by adding the 8-bits value in the accumulator to the 16-bits value in the DPTR register. Neither the accumulator nor the DPTR register are altered. No flags are affected by this instruction.

32. A thread is a light weight process. In the above statement, weight refers to

- (a) time
- (b) number of resources
- (c) speed
- (d) All the above

Ans. (b) : A thread is a light weight process, light weight process means same-resources like memory can be shared between threads.

So answer is (b).

33. The Z-buffer algorithm is used for Hidden surface removal of objects. The maximum number of objects that can be handled by this algorithm shall

- (a) Depend on the application
- (b) be arbitrary no of objects
- (c) Depend on the memory availability
- (d) Depend on the processor

Ans. (b) : In z-buffer-algorithm at each pixel we keep track of the distance to the closest surface that has been drawn so far; and we throw away fragments that are farther away than that distance. The closest distance is stored by allocating an extra value for each pixel, in addition to the red, green and blue color values, which is known as the depth or value. The maximum number of objects that can be handled by this algorithm shall be arbitrary number of objects.

- 34. The power set of $A \cup B$, where $A = \{2, 3, 5, 7\}$ and $B = \{2, 5, 8, 9\}$ is**
(a) 256 (b) 64
(c) 16 (d) 4

Ans. (b) : Because, $A = \{2, 3, 5, 7\}$ and $B = \{2, 5, 8, 9\}$
 $A \cup B = \{2, 3, 5, 7, 8, 9\}$
Power set going to have 2^6 elements which is 64
So the answer is (b)

- 35. In Win 32, which function is used to create Windows Applications?**
(a) Win APP (b) Win API
(c) Win Main (d) Win Void

Ans. (c) : Win main is the function which is provided to create windows application. The user provided entry point for a graphical windows based application.

- 36. Suppose a processor does not have any stack pointer registers, which of the following statements is true?**
(a) It cannot have subroutine call instruction.
(b) It cannot have nested subroutine calls.
(c) Interrupts are not possible.
(d) All subroutine calls and interrupts are possible.

Ans. (d) : It cannot have subroutine call instruction because in nested subroutine calls we used to push old subroutines into stack and pointing most recent call with stack pointer.

- 37. Everything below the system call interface and above the physical hardware is known as**
(a) Kernel (b) Bus
(c) Shell (d) Stub

Ans. (a) : The kernel: consists of everything below the system call interface and above the physical hardware. Provides the file system, CPU-scheduling, memory management and other operating system functions; a large number of functions for one level.

- 38. Which is not the correct statement?**
(a) The class of regular sets is closed under homomorphisms.
(b) The class of regular sets is not closed under inverse homomorphisms.
(c) The class of regular sets is closed under quotient.
(d) The class of regular sets is closed under substitution.

Ans. (b) : Because, The class of regular set is not enclosed under inverse homomorphism but actually regular sets is closed under inverse homomorphism.

- 39. When a programming Language has the capacity to produce new data type, it is called as**
(a) Overloaded Language
(b) Extensible Language
(c) Encapsulated Language
(d) Abstraction Language

Ans. (b) : Because, It supports the style of computer programming that focuses on mechanisms to extend the programming language, compiler & runtime environment. It must be possible to add new keywords, concepts, and structures to the source languages.

40. Which of the following operating system is better for implementing client-server network?

- (a) Windows 95 (b) Windows 98
(c) Windows 2000 (d) All of these

Ans. (c) : Because, windows 2000 is the latest one. window Server 2000 is an operating system for use on both client and server computers.

41. Consider a system having m resources of the same type. These resources are shared by 3 processes A, B and C which have peak demands of 3, 4 and 6 respectively. For what value of m deadlock will not occur?

- (a) 7 (b) 9
(c) 10 (d) 13

Ans. (d) : Because, There are 3 processes A, B and C which have peak demands of 3, 4 and 6 respectively. Maximum number of resources that deadlock can be happen: $\leq (3-1) + (4-1) + (6-1)$

Minimum number of resources that deadlock can not be happen: $m \geq (3-1) + (4-1) + (6-1)+1$

$m \geq 11$

42. The grammar 'G₁'

$S \rightarrow OSO \mid ISI \mid 0 \mid 1 \mid \in$ and the grammar 'G₂' is

$S \rightarrow as \mid asb \mid X, X \rightarrow Xa \mid a.$

Which is the correct statement?

- (a) G₁ is ambiguous, G₂ is unambiguous
(b) G₁ is unambiguous, G₂ is ambiguous
(c) Both G₁ and G₂ are ambiguous
(d) Both G₁ and G₂ are unambiguous

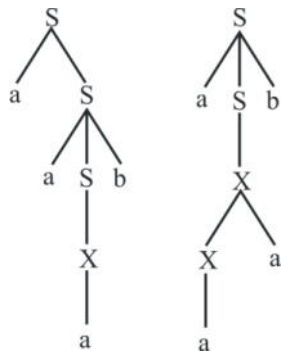
Ans. (b) given grammar G₁:

$s \rightarrow OSO \mid ISI \mid 0 \mid 1 \mid \in$ is unambiguous.

Grammar G₂: $S \rightarrow aS \mid aSb \mid X$

$X \rightarrow Xa \mid a$

For string w=aaab, there are two parse tree:



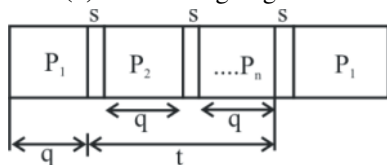
Hence, grammar G₂ is ambiguous grammar,

43. Consider n processes sharing the CPU in round robin fashion Assuming that each process

switch takes a seconds. What must be the quantum size q such that the overhead resulting from process switching is minimized but, at the same time each process is guaranteed to get its turn at the CPU at least every t seconds?

- (a) $q \leq \frac{t - ns}{n - 1}$ (b) $q \geq \frac{t - ns}{n - 1}$
 (c) $q \leq \frac{t - ns}{n + 1}$ (d) $q \geq \frac{t - ns}{n + 1}$

Ans. (a) : according to given data



Therefore, $q(n-1) + n.s \leq t$

$q(n-1) \leq t - n.s$

$$q \leq \frac{t - ns}{n - 1}$$

So answer is (a).

44. The Default Parameter Passing Mechanism is called as

- (a) Call by Value
 (b) Call by Reference
 (c) Call by Address
 (d) Call by Name

Ans. (a) : Call by Value

The call by value method of passing arguments to a function copies the actual value of an argument into the formal parameter of the function. C does not support call by reference, call by address directly. It access call by reference or address by point.

So the default Parameter Passing Mechanism is call by value, Hence the answer is (a).

45. Which of the following regular expression identities are true?

- (a) $(r + s)^* r^* s^*$
 (b) $(r + s)^* = r^* + s^*$
 (c) $(r + s)^* = (r^* s^*)^*$
 (d) $r^* + s^* = r^* s^*$

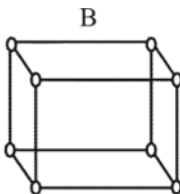
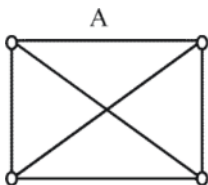
Ans. (c) : $(r + s)^* = (r^* s^*)^*$

Because, Try generating string from both sides.

$$(r+s)^* = (r^*+s^*)^* = (r^*s^*)^* = (r^*s^*)^* = (r^*+s^*)^* = (r + s)^* = a^* (ba^*)^* = b^* (ab^*)^*$$

Hence answer is option (c).

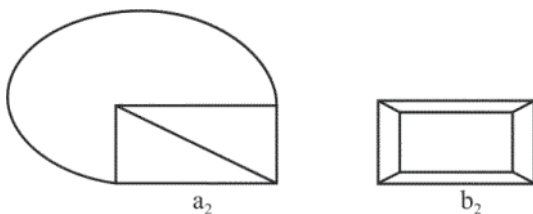
46. Two graphs A and B are shown below : Which one of the following statement is true?



- (a) Both A and B are planar.
- (b) Neither A nor B is planar.
- (c) A is planar and B is not.
- (d) B is planar and A is not.

Ans. (a) : Both A and B are planar.

A planar graph is a graph that can be embedded in the plane. i.e. it can be drawn on the plane in such a way that its edges intersect only at their-end points. It means, it can be drawn in such a way that no edges cross each other.



Hence correct option is (a).

47. The minimum number of states of the non-deterministic finite automation which accepts the language

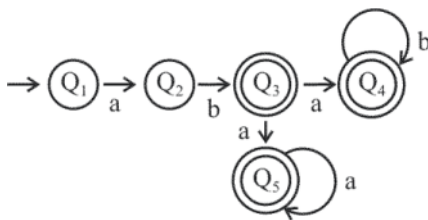
$$\{abab^n \mid n \geq 0\} \cup \{aba^n \mid n \geq 0\}$$

- (a) 3
- (b) 4
- (c) 5
- (d) 6

Ans. (c) : 5

Because, $L = \{abab^n \mid n \geq 0\} \cup \{aba^n \mid n \geq 0\}$

$L = \{ab, aba, abaa, abab, ababb \dots\}$



The minimum number of states of the non deterministic finite automation which accepts the language = 5

Hence option (c) is the answer.

48. Functions defined with class name are called as

- (a) Inline function
- (b) Friend function
- (c) Constructor
- (d) Static function

Ans. (c) : Constructor

Because, constructor has same name as class it

- Constructors don't have return type.
- Constructor is called when object is created.
- If we do not specify constructor then compiler generates a default constructor

49. Let f be the fraction of a computation (in terms of time) that is parallelizable, P the number of processors in the system, and s_p the speed up achievable in comparison with sequential execution – then the s_p can be calculated using the relation :

$$(a) \frac{1}{1-f-f/P}$$

$$(b) \frac{P}{P-f(P+1)}$$

$$(c) \frac{1}{1-f+f/P}$$

$$(d) \frac{P}{P+f(P-1)}$$

Ans. (c) : $\frac{1}{1-f+f/P}$

Because,

- Let execution time without parallelisms 1
- If f is the fraction of parallel computation $(1-f)$ time must be spend for sequential computation.
- f fraction of computation can be distributed among P processors : So it will take f/p time to complete computation
- Total time for computation with parallelism = $1 - f + f/p$ speed up = time without parallelism/time with parallelism. = $1/1-f+f/p$

50. Which of the following definitions generates the same Language as L ,

where $L = \{WW^R \mid W \in \{a,b\}^*\}$

$$(a) S \rightarrow asb|bsa| \in$$

$$(b) S \rightarrow asa|bsa| \in$$

$$(c) S \rightarrow asb|bsa|asa|bsb| \in$$

$$(d) S \rightarrow asb|bsa|asa|bsb$$

Ans. (b) : A palindrome is a word, phrase number or other sequence of characters which reads the same backward or forward, given language

$$S \rightarrow asa|bsa| \in$$

$$L = \{WW^R \mid W \in \{a,b\}^*\}$$

This language accepts even length of palindrome.

The set of strings generated by $L =$

$$\{\epsilon, aa, abba, baab, bbbbbb, bbaabb-\}$$

$\Rightarrow$ We eliminate the (A), (C) because that is not even length palindrome, In option (b) accepts all even length palindrome. Ex : $-\epsilon, aa, bb, aaaa, abba, baab, bbbb$.

$\Rightarrow$ So option (b) is the answer.

51. Suppose there are $\log_n$ sorted lists of $n \log_n$ elements each. The time complexity of producing a sorted list of all these elements is (use heap data structure)

$$(a) O(n \log \log_n)$$

$$(b) \theta(n \log_n)$$

$$(c) \Omega(n \log_n)$$

$$(d) \Omega(n^{3/2})$$

Ans. (a) : $O(n \log \log_n)$

Since we have $\log_n$ lists we can make a min heap of $\log_n$ elements by taking the first element from each of the $\log_n$ sorted lists. Now, we start deleting the min-element from the heap and put the next elements from the sorted list from which that element was added to the heap. In this way each delete and the corresponding

insert will take $O(\log \log_n)$ time as delete in heap size n is $O(\log_n)$ & inserting an element on a heap of size n is also $O(\log_n)$. Now, we have a total of $\log_n \times \frac{n}{\log_n} = n$ elements.
So total time will be $O(n \log \log_n)$

52. Consider the program below in a hypothetical programming language which allows global variables and a choice of static or dynamic scoping int i;

```

program Main( )
{
    i = 10;
    call f ( );
}
procedure f( )
{
    int i = 20;
    call g ( );
}
procedure g( )
{
    print i;

```

Let x be the value printed under static scoping and y be the value printed under dynamic scoping. Then x and y are

- (a) $x = 10, y = 20$ (b) $x = 20, y = 10$
(c) $x = 20, y = 20$ (d) $x = 10, y = 10$

Ans. (a) : In **static scoping**, the scope of an identifier is determined by its location in the code, and since that doesn't change, the scope doesn't either.

In **dynamic scoping**, the scope is determined by sequence of calls that has held to the use of an identifier and since that can be different each time that use is reached, is dynamic.

$x = 10, y = 20$

Since the value of x is based on static scoping in the procedure $g()$ print i will directly look into the global scope & find $i = 10$ which was previously set by $main()$

Since the value of y is based on dynamic point, procedure $g()$ will first look into the function which called its i.e. procedure $f()$ which has a local $i = 20$, which will be taken & 20 will be printed.

53. If the parse tree of a word w generated by a Chomsky normal form grammar has no path of length greater than i , then the word w is of length

- (a) no greater than 2^{i+1} (b) no greater than 2^i
(c) no greater than 2^{i-1} (d) no greater than i

Ans. (c) : no greater than 2^{i-1}

It a parse tree for a word string w is generated by a MF & the parse tree

- has a path length of at most i ,
- then the length of w is at most $2^{(i-1)}$

So answer is (c).

- 54. The Object Modeling Technique (OMT) uses the following three kinds of model to describe a system**
- (a) Class Model, Object Model and Analysis Model.
 - (b) Object Model, Dynamic Model, and Functional Model.
 - (c) Class Model, Dynamic Model and Functional Model.
 - (d) Object Model, Analysis Model and Dynamic Model.

Ans. (b) : Object Modeling technique (OMT) has types of models.

- (i) **Object Model** – It represents the static modeled domain. Main concepts are classes are association with attributes & operations. Aggregation & generalization are predefined relationships.
- (ii) **Dynamic Model** – It represents view on the model main concepts are states, transitions between states, and events to trigger transitions. Action can be modeled as occurring within states generalization and aggregation are predefined relationship.
- (iii) **Functional model** – It handles the process persecuting of the model, corresponding roughly to data flow diagrams. Main concepts are process, data store, and data flow. OMT is a predecessor of the unified modeling language (UML)

- 55. The factors that determine the quality of a software system are**
- (a) Correctness, reliability
 - (b) efficiency, usability, maintainability
 - (c) testability, portability, accuracy, error tolerances, expandability, access control, audit.
 - (d) All of the above

Ans. (d) : All of the above

The ISO 9126 – 1 Software Quality Model identifies 6 main qualities Characteristics namely: Functionality, Reliability, Usability, Efficiency, Maintainability, Portability

- 56. If a relation with a Schema R is decomposed into two relations R_1 and R_2 such that $(R_1 \cup R_2) = R$ then which one of the following is to be satisfied for a lossless joint decomposition ($\rightarrow$ indicates functional dependency)**
- (a) $(R_1 \cap R_2) \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$
 - (b) $R_1 \cap R_2 \rightarrow R_1$
 - (c) $R_1 \cap R_2 \rightarrow R_2$
 - (d) $R_1 \cap R_2 \rightarrow R_1$ and $R_1 \cap R_2 \rightarrow R_2$

Ans. (a) : Let $R(A, B, C)$ and B is the key
 Let $R_1 = (A, B)$ and $R_2 = (B, C)$ So $R_1 \bowtie R_2 = R_1$
 Definition of lossless D : says that if the intersection of
 two relation has a common attribute and that is a key
 in any of the relation then it is lossless
 Now $R_1 \cap R_2 = B \dots$ AND $B \rightarrow R_1$
 So lossless and it can also be stated that
 $R_1 \cap R_2 = B \dots$ AND $B \rightarrow R_2 \dots \dots \dots$
 So lossless therefore if we have any one of the above
 then it is lossless

57. Given the following statements :

- (i) Recursive enumerable sets are closed under complementation.
- (ii) Recursive sets are closed under complementation.

Which is/are the correct statements?

- (a) only (i) (b) only (ii)
(c) both (i) and (ii) (d) neither (i) nor (ii)

Ans. (b) : only (ii)
Recursive enumerable sets are not closed under complementation.
Recursive languages are closed under complement.
So the answer is (b).

58. Skolmization is the process of

- (a) bringing all the quantifiers in the beginning of a formula in FDL.
- (b) removing all the universal quantifiers.
- (c) removing all the existential quantifiers.
- (d) all of the above

Ans. (c) : Because, Skolemization is the process of removing all the existential quantifiers. It is one of the steps in conversion of first order logic sentences into its equivalent CNF (conjunctive normal form)

59. Which level of Abstraction describes how data are stored in the data base?

- (a) Physical level (b) View level
(c) Abstraction level (d) Logical level

Ans. (a) : Physical level is the lowest level of data abstraction. This level describes how data is actually stored in the physical memory or database.

60. The transform which possesses the "multi-resolution" property is

- Fourier transform
- Short-time-Fourier transform
- Wavelet transform
- Karhunen-Loere transform

Ans. (c) : A multi resolution analysis (MRA) or multi scale approximation (MSA) is the design method of most of the practically relevant discrete wavelet transform (DWT) & the justification for the algorithm of the fast wavelet transform (FWT)

Wavelet transforms are broadly divided into three classes continuous, discrete and multi-resolution based.

61. Which one is a collection of templates and rules?

- (a) XML
- (b) CSS
- (c) DHTML
- (d) XSL

Ans. (d) : XSL stands for extensible style sheet language and is a style sheet language for XML documents. An XSL style sheet is like with CSS, a file that describes how to display an XML document of a given type.

62. A program P calls two subprograms P₁ and P₂. P₁ can fail 50% times and P₂ 40% times. Then P can fail

- (a) 50%
- (b) 60%
- (c) 10%
- (d) 70%

Ans. (d) : Program P fails when either P₁ fails or P₂ fails i.e. failure of P₁ failure of P₂.

But this will also contain the case when both P₁ & P₂ fails at same time i.e. failure of P₁ failure of P₂, since this case will be already be counted on (P₁ + P₂)

Therefore, our final answer will be failure of P₁ + failure of P₂ – (failure of P₁ ∩ failure of P₂)

$$\Rightarrow \left(\frac{50}{100}\right) + \left(\frac{40}{100}\right) - \left(\frac{50}{100} \times \frac{40}{100}\right)$$

$$\Rightarrow \left(\frac{90}{100}\right) - \left(\frac{2000}{10000}\right)$$

$$= \left(\frac{20}{100}\right) - \left(\frac{200}{1000}\right) = \left(\frac{70}{100}\right) = 70\%$$

63. Third normal form is based on the concept of – _____.

- (a) Closure Dependency
- (b) Transitive Dependency
- (c) Normal Dependency
- (d) Functional Dependency

Ans. (b) : Transitive Dependency

Third normal form (3NF) is a normal used in data base normalization.

- The relation R (table) is in second normal form 2MF
- Every non-prime attribute of R is not-transitively dependent on every key of R.

A non-prime attribute of R is an attribute that does not belong to any candidate key of R.

A transitive dependency is a functional dependency in which $X \rightarrow Z$ indirectly, by virtue of

$X \rightarrow Y$ & $Y \rightarrow Z$ (where it is not the case that $Y \rightarrow X$)

So the option is (b).

64. If the Fourier transform of the function f(x, y) is F(m, n), then the Fourier transform of the function f(2x, 2y) is :

- (a) $\frac{1}{4} F\left(\frac{m}{2}, \frac{n}{2}\right)$
- (b) $\frac{1}{4} F(2m, 2n)$

$$(c) \frac{1}{4}F(m, n)$$

$$(d) \frac{1}{4}F\left(\frac{m}{4}, \frac{n}{4}\right)$$

Ans. (a) : To determine the factor by which the frequency change, recalls

$$f(\omega) = \int_{-\infty}^{\infty} f(x) e^{2\pi i x \omega} dx$$

Replace x with 2x. Then

$$f(\omega) = \int_{-\infty}^{\infty} f(2x) e^{4\pi i x \omega} dx$$

Let, 2x = u,

$$\text{Then, } \frac{du}{dx} = 2 \text{ or } dx = \frac{du}{2}$$

Substituting gives :

$$f(\omega) = \frac{1}{2} \int_{-\infty}^{\infty} f(u) e^{2\pi i u \omega} du$$

But this integral is the same as the defining integral for the Fourier transform for f(x) with x' replaced with u's. Therefore,

$$f(\omega) = \frac{1}{2} F(\omega')$$

But we know $\omega' = m/2$

$$\text{so } F(\omega) \text{ for } f(2x) \text{ is } \frac{1}{2} F\left(\frac{m}{2}\right)$$

By similar argument,

$$\text{for } f(3y), F(\omega) = \frac{1}{2} F\left(\frac{m}{2}\right)$$

Combining the functions gives the Fourier transform,

$$\text{for } f(2x, 2y) \text{ as } \frac{1}{2} F\left(\frac{m}{2}, \frac{n}{2}\right)$$

65. establishes information about when, why and by whom changes are made in a software.

- Software Configuration Management.
- Change Control.
- Version Control.
- An Audit Trail.

Ans. (d) : An Audit Trail.

A record showing who has accessed a computer system and what operations he or she has performed during a given period of time. Audit trails are useful both for maintaining security and for recovering lost transactions.

66. Match the following with respect to HTML tags and usage.

List-I

- CITE
- EM
- VAR
- SAMP

List-II

- Italic representation
- Represents output from programmes
- Represents to other source
- Argument to a programme

Codes :

	a	b	c	d
(a)	3	1	4	2
(b)	2	3	1	4
(c)	4	2	3	1
(d)	1	3	4	1

Ans. (a) :

- (a) CITE – defines the title of a work.
- (b) EM – defines emphasized text.
- (c) VAR – defines the variable
- (d) SAMP – defines the sample output from a computer system.

So answer is (a).

67. An expert system shell is an expert system without

- (a) domain knowledge
- (b) explanation facility
- (c) reasoning with knowledge
- (d) all of the above

Ans. (a) : domain knowledge: A shell is an expert system without a knowledge base. A shell furnishes the expert system developer with the inference engine, user interface, and the explanation and knowledge acquisition facilities.

68. An example of a dictionary-based coding technique is

- (a) Run-length coding
- (b) Huffman coding
- (c) Predictive coding
- (d) LZW coding

Ans. (d) : LZW coding

Option A, B, C all are statistical based coding techniques. In statistical based coding techniques we will encode text according to the frequent occurrence of the alphabets.

Hence option (d) : **LZW Coding** is universal lossless data compression algorithms is the algorithm the widely used in the GIF image format. It's an example for dictionary based compression algorithm. It uses a dictionary to code the input text

69. Which is the method used to retrieve the current state of a check box?

- (a) get State ()
- (b) put State ()
- (c) retrieve State ()
- (d) write State ()

Ans. (a) : get State ()

Get state method returns the current state. Java.lan. Thread. get state () method returns the state of this thread. It is designed for use in monitoring of the system state, not for synchronization control.

So get state () can retrieve the current state of check box.

70. Referential integrity is directly related to

- (a) Relation key
- (b) Foreign key
- (c) Primary key
- (d) Candidate key

Ans. (b) : Foreign key: is a field in one table that uniquely identifies a row of another table or the same table. This is sometimes called a referencing to any other data.

71. You are given four images represented as

$$I_1 \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, I_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

$$I_3 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix},$$

The value of entropy is maximum for image

- (a) I_1 (b) I_2
(c) I_3 (d) I_4

Ans. (c) : I_3

Entropy of image represents the randomness in pixels. In I_2 all pixels are same entropy is 0 : in I_3 2 pixels are same other 2 different in I_4 , 3 are same So most random is in I_3

72. A cryptarithmic problem of the type

SEND
+ MORE
MONEY

Can be solved efficiently using

- (a) depth first technique
(b) breadth first technique
(c) constraint satisfaction technique
(d) bidirectional technique

Ans. (c) : Constraint satisfaction technique is a genre of mathematical puzzles in which the digits are replaced by letters of the alphabet or other symbols. It has one to one correspondence.

In this cryptarithmic problem there are some constraints like.

- (1) no two alphabets can have same number
- (2) every number can have values in the range 0 – 9

By considering these constraints we will solve the above problem which satisfy these constraints like M will be 1 as it is carry & so on.

73. Match the following :

List-I

List-II

- | | |
|----------------------------|---|
| a. Supervised learning | 1. The decision system receives rewards for its action at the end of a sequence of steps. |
| b. Unsupervised learning | 2. Manual labels of inputs are not used. |
| c. Re-inforcement learning | 3. Manual labels of inputs are used. |
| d. Inductive learning | 4. System learns by example |

Codes :

	a	b	c	d
(a)	1	2	3	4
(b)	2	3	1	4
(c)	3	2	4	1
(d)	3	2	1	4

Ans. (d) : Reinforcement learning is an area of machine learning inspired by behaviorist psychology concerned with how software agent ought to take actions in an environment so as to maximize some notion of cumulative reward C-1

Inductive learning uses learn by example system.

Supervised learning is the machine learning task of inferring a function from labeled training data. There may manual labels of inputs are used.

Unsupervised learning is a type of machine learning algorithm used to draw inferences from data sets consisting of input data without labeled responses.

Hence answer is (d)

74. A algorithm is guaranteed to find an optimal solution if

- (a) h' is always 0.
- (b) g is always 1.
- (c) h' never overestimates h .
- (d) h' never underestimates h .

Ans. (c) : h' never overestimates h

A^* is a computer algorithm that is widely used in path finding & graph traversal.

A^* is an informed search algorithm, or a best-first search, meaning that it solves problems by searching among all possible paths to the solution for the one that incurs the smallest cost.

A^* algorithm is guaranteed to find an optimal solution if h' never overestimates h .

75. Let $\theta(x, y, z)$ be the statement " $x + y = z$ " and let there be two quantifications given as

(i) $\forall x \forall y \exists z \theta(x, y, z)$

(ii) $\exists z \forall x \forall y \theta(x, y, z)$

Where x, y, z are real numbers. Then which one of the following is correct?

- (a) (i) is true and (ii) is true.
- (b) (i) is true and (ii) is false.
- (c) (i) is false and (ii) is true.
- (d) (i) is false and (ii) is false.

Ans. (b) : (i) is true and (ii) is true.

$x + y = z$ for all x for all y there exist some z which will satisfy this equation as e.g. $x = 4351$ $y = 1111$ then some $z = 5462$ is there and so on

for some z say $z = 100$ there do not exist all x and all y (there exist only some x, y) which satisfies this equation hence II is false.

So answer is (b).